



3593 - Towards Solving the Stellar Inhomogeneity Contamination of Exoplanet Transmission Spectra Problem with Star Spot and Faculae Spectra

Cycle: 2, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Stellar surface inhomogeneities, such as spots, faculae, and magnetic network contaminate and, in some cases, overwhelm the planetary signal in transmission spectra (see, e.g. a detailed review by Rackham et al. 2022). In particular the lofty goal of analyzing Earth-size planet atmospheres in the habitable zones of M dwarf stars may be prevented until we can disentangle the star surface inhomogeneity from the exoplanet atmosphere signal, e.g., Trappist-1. So far, astronomers have tried to remove the magnetic contamination of transmission spectra by using non-magnetic stellar models—a hotter star to model faculae and a cooler star to model star spots. Yet this approach dramatically fails to reproduce the complex

wavelength dependence of magnetic contamination (Witzke et al. 2022). Thus, a more comprehensive approach based on realistic magnetohydrodynamics (MHD) simulations of stellar atmospheres and magnetic features within these atmospheres is needed. The magnetohydrodynamics simulations have reached a high degree of maturity and have already revolutionized solar and stellar physics. Now they are ready to address the big challenge facing exoplanetary science and allowing us to overcome the barrier imposed by magnetic contamination. We propose to generate a “library” of star spot and star faculae spectra of M and K stars for the community to use in furthering understanding contamination of transmission spectra light curves for atmospheres of exoplanets and, in particular, of rocky exoplanets transiting M dwarf stars.

OBSERVING DESCRIPTION

N/A This is an AR theory proposal.